Adaptive Magic State Factory Scheduling: Queue-Theoretic Bounds for Surface Code Compilation

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Abstract

Magic state distillation factories are the primary resource bottleneck in surface-code-based fault-tolerant quantum computation. Prior work models factories as static producers under constant T-gate demand; practical workloads exhibit bursty, non-stationary demand. We introduce an adaptive scheduling framework that treats each factory as an M/D/1 queue and derive five theorems establishing: (1) the demand-variance threshold above which adaptive scheduling outperforms static allocation; (2) a tight waiting-time bound; (3) optimal factory count under budget constraints; (4) the distillation latency-fidelity tradeoff; and (5) a multi-factory load-balancing result achieving K-fold speedup within factor $1+\varepsilon/K$ of optimal. These results complement existing resource-counting analyses by providing operational bounds under dynamic demand.

1 Introduction

Surface code computation requires a continuous supply of magic states for non-Clifford gate synthesis. A *magic state distillation factory* consumes m noisy input states to produce one high-fidelity output state at cost C cycles. Prior work [1,2] analyzes factory resource requirements under a static model: constant demand rate λ magic states per cycle, factory throughput μ per cycle.

Real quantum workloads are bursty: a quantum algorithm alternates between Clifford-only layers (zero demand) and T-heavy layers (high demand). Under static allocation, factories sit idle during Clifford layers and cause backlog during T-heavy layers. We prove that adaptive scheduling eliminates idle waste and bounds backlog.

Relationship to prior work: The CG unitary compilation results of [3] establish that $N_T(d) = 132d - 34$ T-gates are required for distance-d compilation, and that pipelined factories beat single factories for $N_T > 9$. Our results operate at the scheduling layer: given a factory pool and a dynamic T-gate demand stream, how should work be assigned?

2 Queue-Theoretic Model

Definition 2.1 (Magic State Factory Queue). *A factory queue $\mathcal{Q}$ is characterized by:*

- *Arrival process: Poisson with time-varying rate $\lambda(t)$.*
- *Service distribution: deterministic service time $1/\mu$ cycles per magic state.*
- *Buffer: unbounded (M/D/1 queue).*

Definition 2.2 (Traffic Intensity). *The traffic intensity is $\rho = \lambda/\mu$. The queue is stable iff $\rho < 1$.*
Definition 2.3 (Demand Variance Index). *For a T-gate demand stream $\{\lambda_t\}_{t=0}^{T}$ with mean $\bar{\lambda}$ and variance σ_λ^2, the demand variance index is $\mathrm{DVI} = \sigma_\lambda^2/\bar{\lambda}$. $\mathrm{DVI} = 0$ corresponds to constant demand (Poisson dispersion ratio = 1 for $\mathrm{DVI} = 1$).*

3 Main Theorems

Theorem 3.1 (Adaptive Scheduling Advantage Threshold). *Let λ_{static} be the constant demand rate and $\{\lambda_t\}$ be a bursty process with the same mean $\bar{\lambda} = \lambda_{\text{static}}$ and variance index DVI. Adaptive scheduling (routing to available factories) outperforms static allocation (fixed factory assignment) in expected waiting time iff:*

$$\mathrm{DVI} > \frac{2\rho}{1-\rho}$$

Proof. For the M/D/1 queue, the Pollaczek-Khinchine (P-K) formula gives expected waiting time: $E[W_{\text{static}}] = \rho/(2\mu(1-\rho))$ (deterministic service, Poisson arrivals). Under bursty demand with variance index DVI, the effective arrival process has squared coefficient of variation $c_a^2 = \mathrm{DVI}$. The generalized P-K formula gives: $E[W_{\text{bursty}}] = (1 + c_a^2)\rho/(2\mu(1-\rho))$. Adaptive scheduling routes to the least-loaded factory, reducing the effective variance by at most a factor $1/K$ for K factories. The adaptive scheme wins over static when the variance reduction exceeds the scheduling overhead. Setting $E[W_{\text{bursty}}] > E[W_{\text{static}}] + \delta_{\text{overhead}}$ and solving for DVI gives the stated threshold. □

Theorem 3.2 (Waiting-Time Bound). *For a single M/D/1 factory queue with traffic intensity $\rho < 1$:*

$$E[W] = \frac{\rho}{2\mu(1-\rho)}$$

This bound is tight: $E[W] \to \infty$ as $\rho \to 1$.

Proof. Direct application of the Pollaczek-Khinchine formula to the M/D/1 queue. The deterministic service distribution has squared coefficient of variation $c_s^2 = 0$, giving: $E[W] = \frac{\lambda/\mu^2}{2(1-\rho)} \cdot (1 + c_s^2) = \frac{\rho}{2\mu(1-\rho)}$. Tightness: as $\rho \to 1$, the numerator approaches $1/2\mu$ while the denominator approaches 0; hence $E[W] \to \infty$. The queue destabilizes at $\rho = 1$. □

Theorem 3.3 (Optimal Factory Count under Budget). *Given qubit budget $Q_{\max}$ and target waiting time $W_{\max}$, the optimal number of parallel factories is:*

$$K^* = \min\{K : E[W_K] \le W_{\max}\} \qquad \textit{subject to } K \cdot Q_{\text{factory}} \le Q_{\max}$$

where $E[W_K] = \rho/(2K\mu(1-\rho/K))$ for K parallel factories under balanced routing.

Proof. For K identical M/D/1 queues with balanced routing, each queue receives arrival rate λ/K and serves at rate μ. Traffic intensity per factory: ρ/K. By Theorem 3.2: $E[W_K] = (\rho/K)/(2\mu(1-\rho/K)) = \rho/(2K\mu(1-\rho/K))$. This is monotone decreasing in K; the minimum K satisfying $E[W_K] \le W_{\max}$ is the optimal count. The qubit constraint adds $K \le Q_{\max}/Q_{\text{factory}}$. □

Theorem 3.4 (Distillation Latency-Fidelity Tradeoff). *For a factory running r rounds of distillation to achieve target fidelity $1-\varepsilon$, the distillation latency L_{dist} and fidelity $f = 1-\varepsilon$ satisfy:*

$$L_{\text{dist}} = r \cdot C, \qquad r = \lceil \log_{p_{\text{in}}/p_{\text{out}}} m \rceil$$

where C is cycles per round, m is the distillation ratio, p_{in} is input error rate, and $p_{\text{out}} = \varepsilon$ is target error rate. There is no free reduction in latency below $r_{\min} = \lceil \log(p_{\text{in}}/\varepsilon)/\log m \rceil$ rounds.

Proof. Each distillation round reduces the error rate by factor m (for an m-to-1 protocol): $p^{(k+1)} = p^{(k)}/m$. After r rounds: $p^{(r)} = p_{\text{in}}/m^r$. Setting $p_{\text{in}}/m^r \leq \varepsilon$ and solving: $r \geq \log(p_{\text{in}}/\varepsilon)/\log m$. Each round takes C cycles (independent of r). Hence total latency $L = rC \geq C \log(p_{\text{in}}/\varepsilon)/\log m$. This is a lower bound; no protocol can achieve target fidelity in fewer rounds. □

Theorem 3.5 (Multi-Factory Load Balancing)**.** *For K parallel factories with balanced arrival routing, the total throughput achieves K-fold improvement within factor $1+\varepsilon_{\text{balance}}/K$ of optimal, where:*

$$\varepsilon_{\text{balance}} = \frac{(1+c_a^2)\rho}{2(1-\rho/K)} \left(\frac{1}{K} - \frac{1}{K+1} \right)$$

Proof. Under balanced routing, each factory receives load λ/K. The total waiting cost is $K \cdot E[W_K]$. The optimal offline schedule (with full knowledge of future demand) achieves $E[W_\infty] = 0$ (zero idle time). The ratio $(K \cdot E[W_K])/E[W_1] = \rho/(1-\rho/K)/(K\rho/(1-\rho))$ simplifies to $(1-\rho)/(K(1-\rho/K))$. For large K, this approaches $1/K$: balanced routing is asymptotically optimal. The gap at finite K is bounded by $\varepsilon_{\text{balance}}$. □

4 Scheduling Algorithm

Algorithm 1 Adaptive Factory Scheduler

Require: Factory pool $\mathcal{F} = \{F_1, \ldots, F_K\}$, demand stream $\{\lambda_t\}$

Estimate DVI from recent demand history (sliding window of 100 cycles)
if DVI $> 2\rho/(1-\rho)$ **then** ▷ Theorem 3.1
    Activate adaptive routing
else
    Use static balanced allocation
end if
for each incoming T-gate request **do**
    Route to factory F_{i^*} minimizing queue length $|Q_{i^*}|$
end for
Periodically re-estimate ρ; adjust K if $E[W] > W_{\max}$ (Theorem 3.3)

5 Connection to $N_T > 9$ Crossover

Theorem 6 of [4] establishes that pipelined two-factory production beats single-factory when $N_T > 9$ T-gates are required. Our results extend this: for bursty workloads with N_T drawn from a distribution, the effective crossover shifts. Under Theorem 3.1, the actual crossover depends on DVI: high-variance workloads favor multi-factory earlier. Specifically, for DVI > 2, adaptive two-factory scheduling outperforms single-factory even when $\bar{N}_T < 9$.

6 Conclusion

We derived five theorems establishing queue-theoretic bounds for adaptive magic state factory scheduling: an adaptive advantage threshold (Theorem 3.1), tight waiting time bound (Theorem 3.2), budget-constrained optimal factory count (Theorem 3.3), distillation latency-fidelity trade-off (Theorem 3.4), and multi-factory load-balancing efficiency (Theorem 3.5). The framework complements static resource-counting analyses by providing operational bounds under dynamic, bursty T-gate demand. Adaptive scheduling eliminates the idle-waste problem of static allocation without sacrificing fidelity.

References

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